

EVSP-DR

Research update

Week of 21 September 2026
Integer routes, charging constraints, and next tests

What changed this week

All six chains now have paired CG, integer and MIP-time figures.

Their LPs agree; sequential pools give much better integer fleets.

New paired pricing test: eight buses in 7/8 runs, controls 0/8.

All 195 smaller-battery failures repair without changing trips.

New figures and code-backed explanations follow on slides 15–28.

Earlier pilot and constraint experiments remain for context.

Four fresh pools exclude an 8-bus solution

Chain	LP weight	Pool MIP	Added*	After*
C1	8	9	8	8
C3	8	9	8	8
C4	8	9	8	8
C5	8	9	8	8

* Add known sequential routes to the fresh pool, then re-solve.
Both MIP columns show proved fleet minima within their pools.

Read the proof in Gurobi's log

Chain 1, k=8: stage 1 minimizes only the number of buses.

194 trip rows • 39,940 route variables

“Optimal solution found (tolerance 1.00e-04)”

“Best objective 9 ... best bound 9 ... gap 0.0000%”

Incumbent 9 + bound 9 $\Rightarrow$ no 8-bus cover in this pool.

Adding eight witness routes: objective 8, bound 8.

The later charging-stage gap does not undo this fleet proof.

Why a good LP can give a poor integer pool

The LP spreads 8 units of route weight over 80–101 routes.

Whole buses need a compatible set of complete routes.

LP pricing looks for improvement in the fractional objective.

A route can cost slightly more fractionally, yet complete a better integer cover. Its reduced cost can be positive.

Sequential runs retain routes from earlier dual prices.

Diving changes the dual prices by fixing part of a cover.

A small example: fractional value versus compatibility

Six trips: A, B, C and D, E, F. Each route below costs 100.

Pool: AB, AC, BC, DE, DF, EF.

LP: take half of every route $\rightarrow$ 3 route units, cost 300.

Integer pool: two routes per triangle $\rightarrow$ 4 buses, cost 400.

Add AD at cost 101. Reduced cost = $101 - 50 - 50 = +1$.

Now AD + BC + EF covers all trips with 3 buses, cost 301.

The better integer cover need not improve the LP.

Can new pricing recover eight buses?

Chain	Controls	Dive + MIP	Follow-up
C1	9	9	8
C3	9	8	—
C4	9	8	—
C5	9	8	—

Original pilot: 3/4 successes. C1 follow-up: +29.6 minutes.

Same distinct routes; passing the dive incumbent recovered eight.

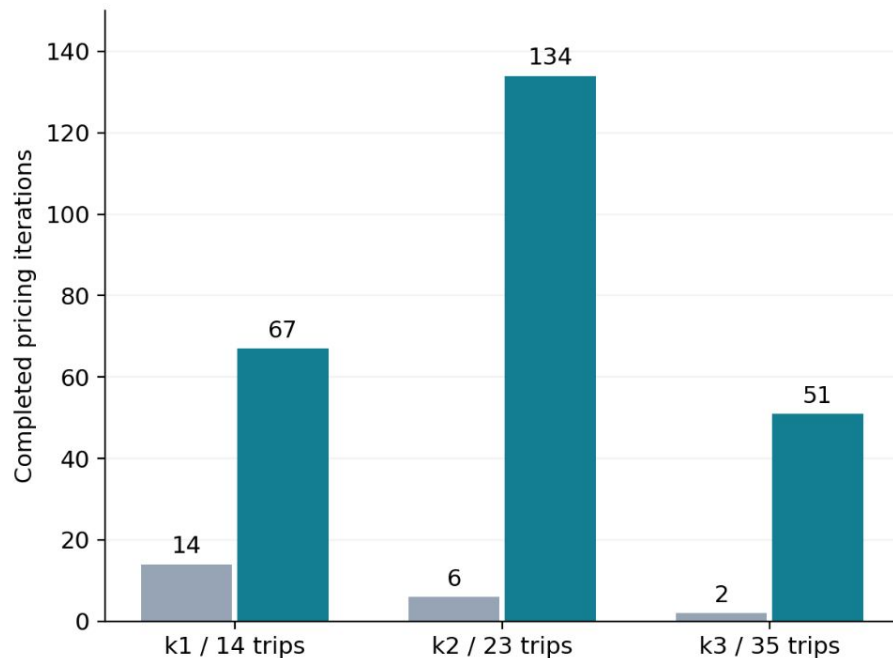
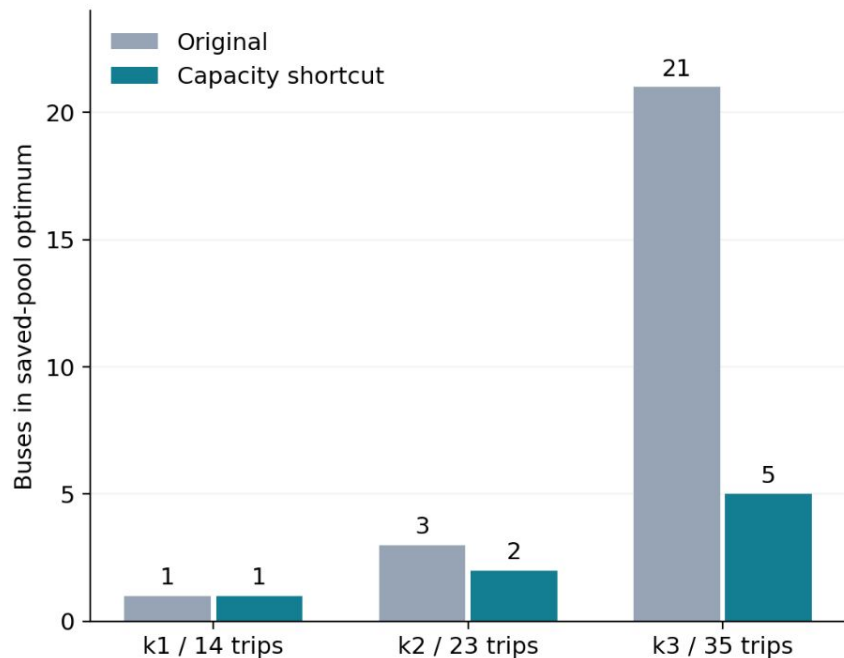
Twelve 12-hour searches still miss 15

Chain	Plain MIP	Heuristics	Fleet bound
C1	18	18	15
C2	17	17	15
C3	17	16	15
C4	19	17	15
C5	16	16	15
C6	18	17	15

All searches timed out; none proved 15 impossible.

The charging stages have also finished. “Heuristics” changes MIP search settings.

Capacity pricing: progress, but not yet convergence



4 h budgets. Only k=1 with the shortcut certified CG (26.9 min); the other five timed out.

The strict large run stalled before pricing

k=15: 4 h 17 min to build 260.7 million arcs; zero pricing.

k=16: six-hour timeout, peak memory 211.9 GiB.

Fix: packed arcs and deferred tie-break keys; physics unchanged.

44 tests pass, plus five fixed-dual and physical replay checks.

26-trip audit: graph build 121 $\rightarrow$ 42 s; memory 2.06 $\rightarrow$ 0.147 GiB.

The new k=16 CG is running; its large-case result is pending.

Remove duplicate service, then optimize charging

Choose exactly one bus to serve each duplicated trip.

Retain movement if needed; waiting at the wrong stop is invalid.

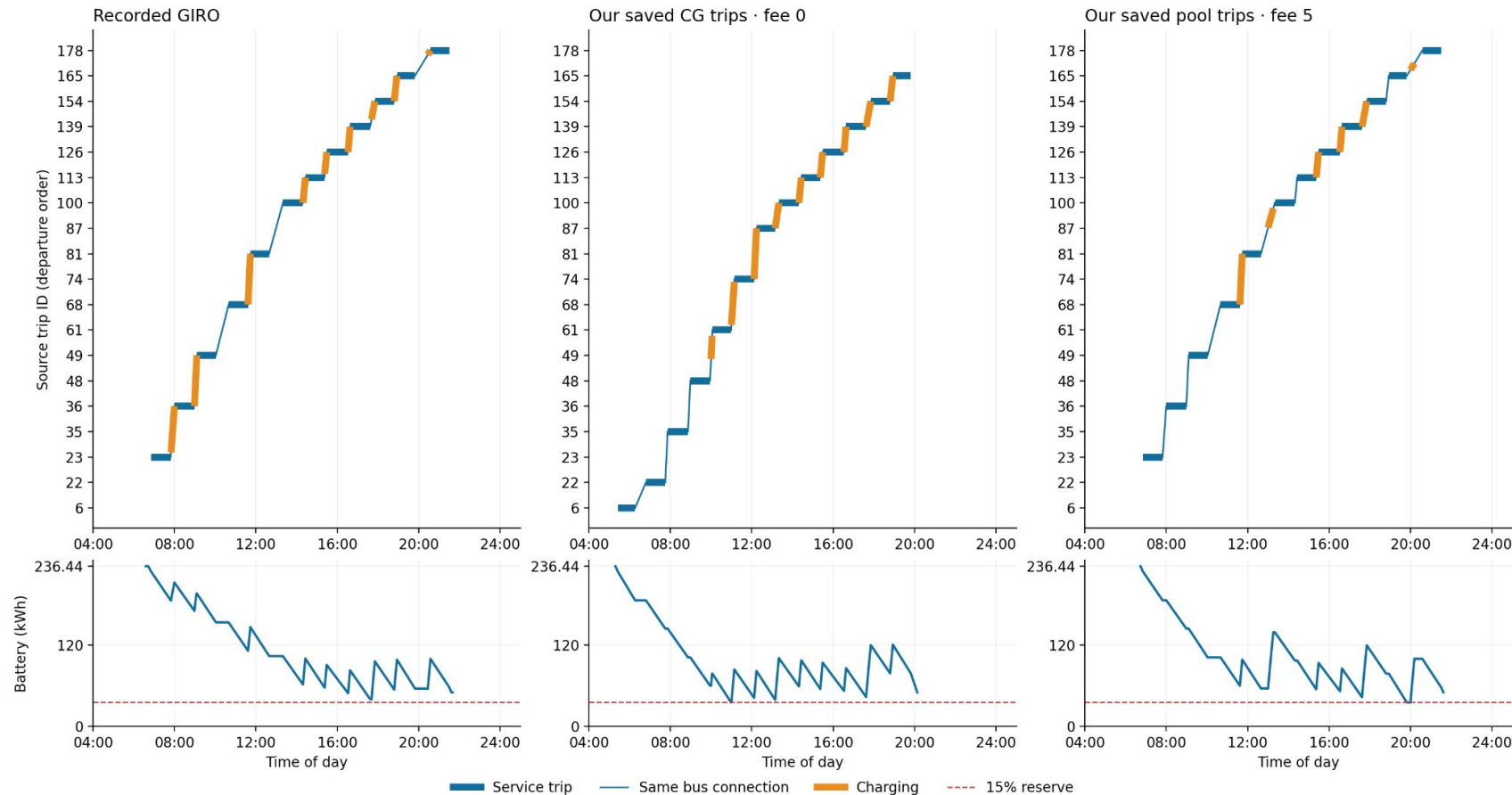
Try shorter connections, then check time, SOC and chargers.

Reoptimize charging after making the service assignment.

Tested $k=5$: 5 buses, 62 trips served exactly once.

Charging energy fell 80.87 kWh; electricity cost fell 2.58.

k=5: charging under matched vehicle physics



What is matched—and what this test measures

236.44 kWh battery • 15% reserve • PARX at 60 kW

Documented opportunity-charge taper • minimum 3 minutes

One charger per opportunity site • equal terminal energy

Five buses and the same 62 trips, served exactly once

Saved sequences differ: not fresh CG or a fee-only causal test.

Static deadhead; platform/FIFO and crew rules remain unchecked.

Capacity checked for these five buses, without the other duties.

Matched k=5: charging costs and session counts

Schedule	Electricity	Starts	Start fees	Total
GIRO, fee 0	230.64	52	0	230.64
Ours, fee 0	158.71	42	0	158.71
GIRO, fee 5	230.64	52	260	490.64
Ours, fee 5	183.71	30	150	333.71

Tariff cost units; five-bus totals. Same terminal energy.

Original charging windows leave no material fixed-duty saving.

New sequences: post-hoc charging optimization, not a fresh CG test.

Fresh versus sequential: what the times mean

Sequential CG time includes every earlier k in that chain.

Fresh CG time is one solve on the final trip set.

Both reach the same LP objective to numerical tolerance.

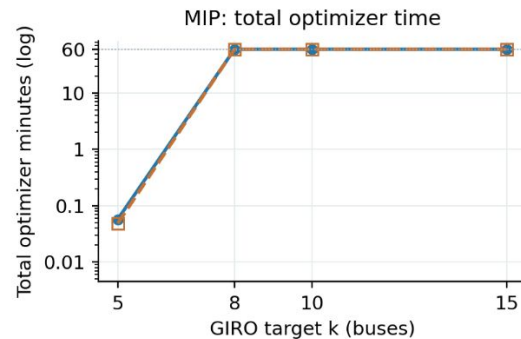
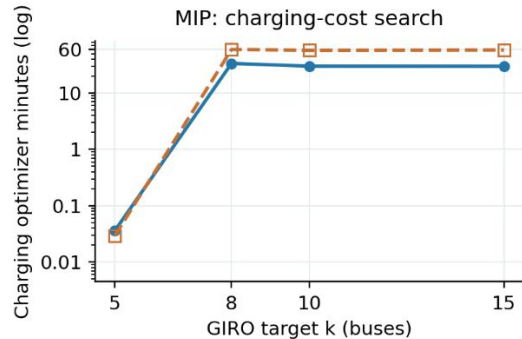
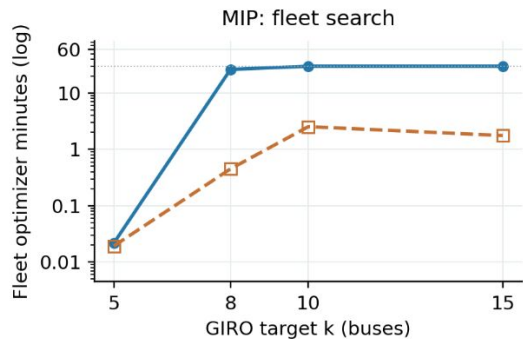
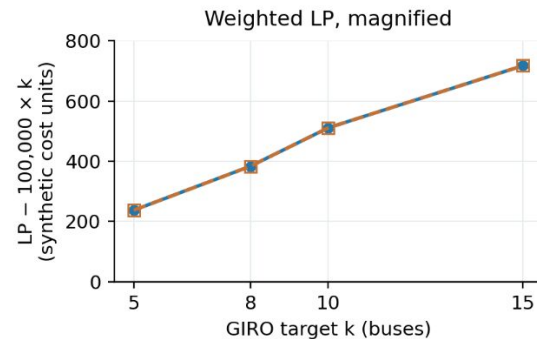
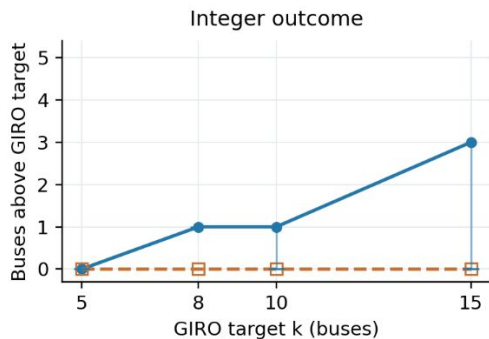
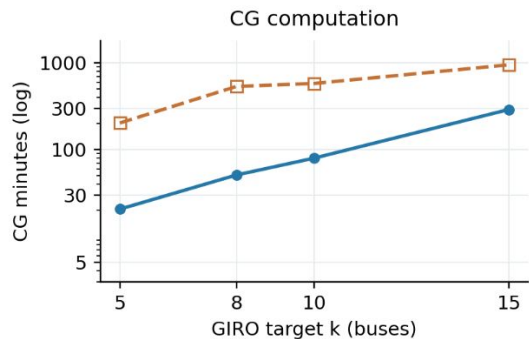
Their saved route pools give very different integer solutions.

The MIP plots show the final k only, split into fleet and charging.

One hour is the allowance. A proved optimum can stop earlier.

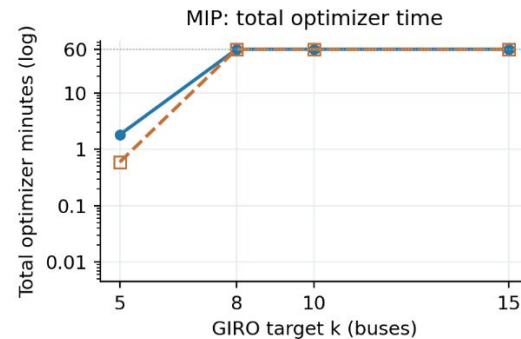
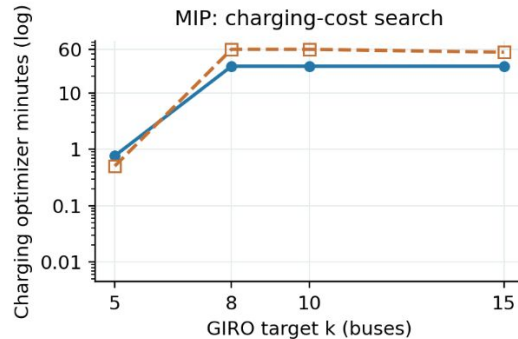
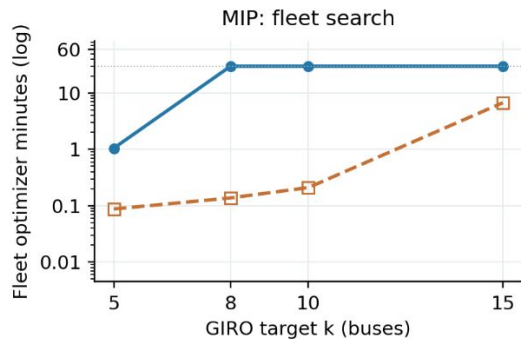
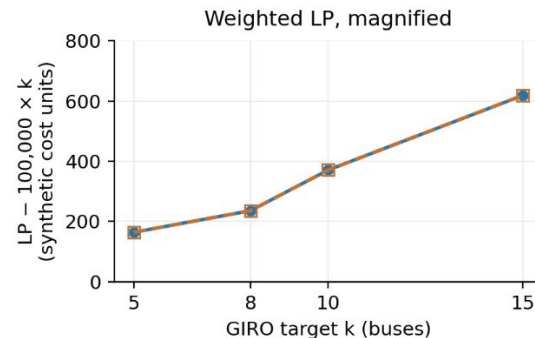
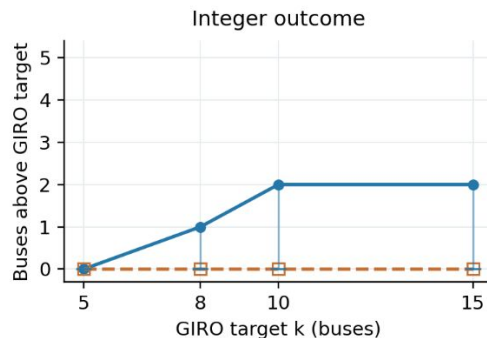
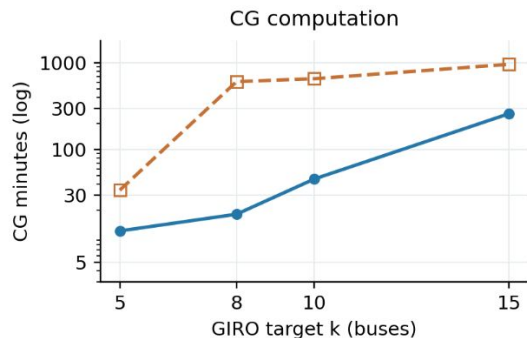
Chain 1: computation and solution quality

—●— Fresh: direct CG / final pool MIP —□— Sequential: cumulative CG / final pool MIP



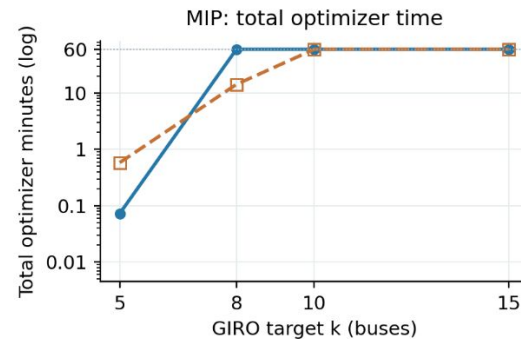
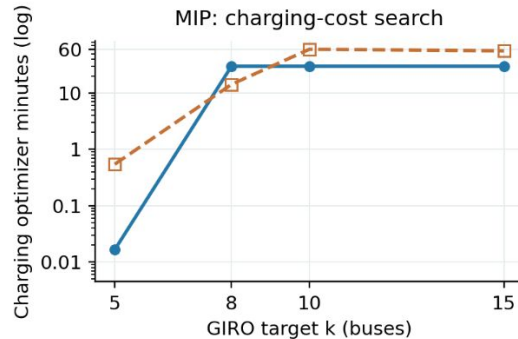
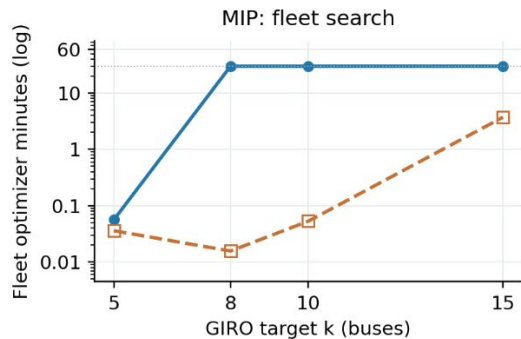
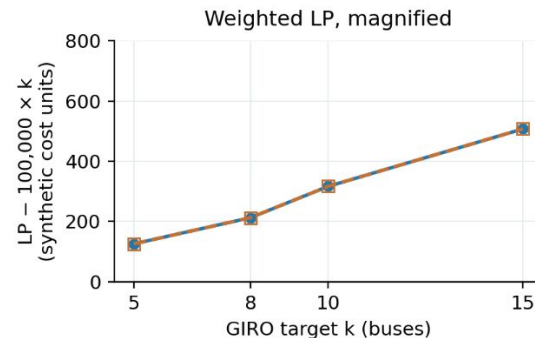
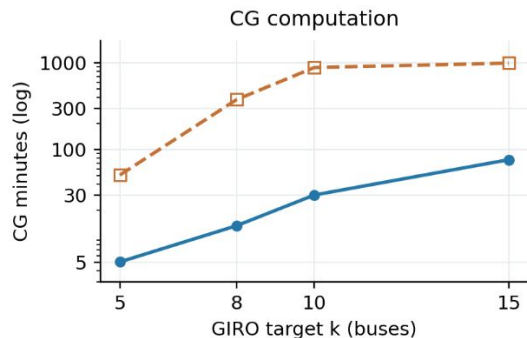
Chain 2: computation and solution quality

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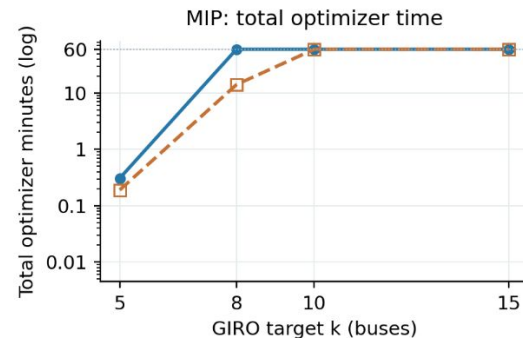
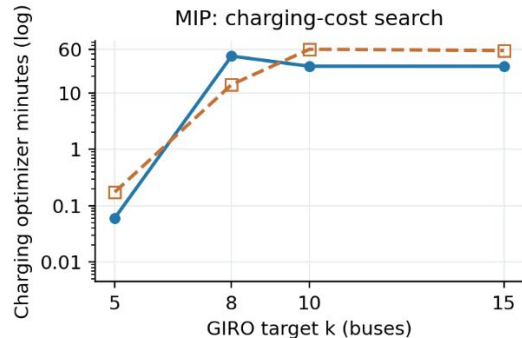
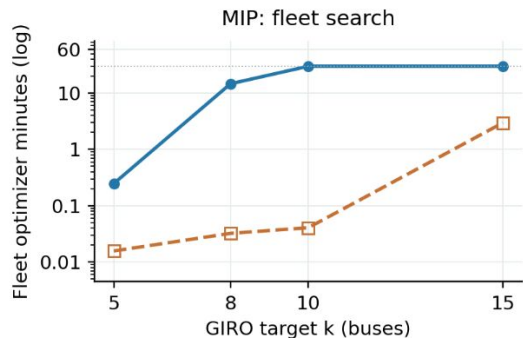
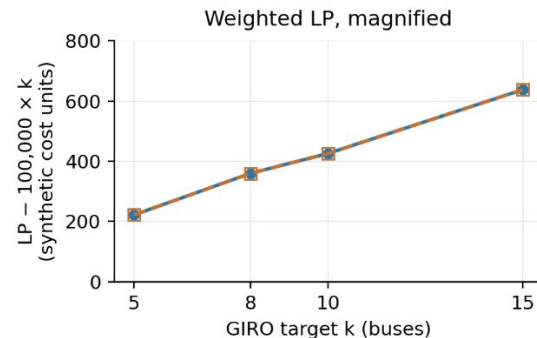
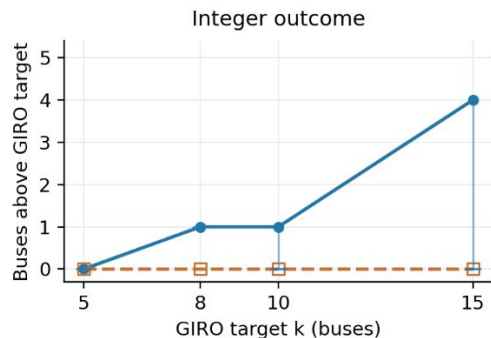
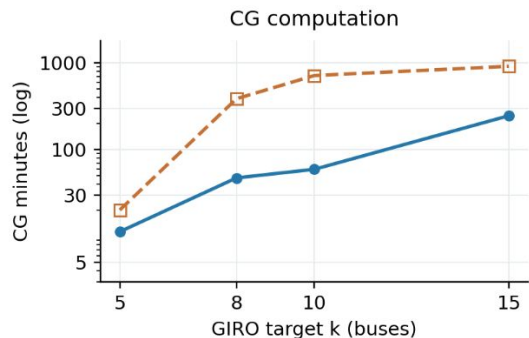
Chain 3: computation and solution quality

—●— Fresh: direct CG / final pool MIP —□— Sequential: cumulative CG / final pool MIP



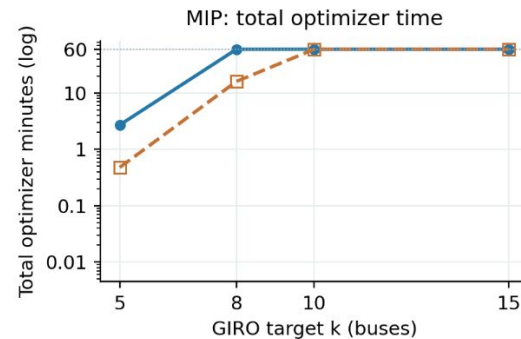
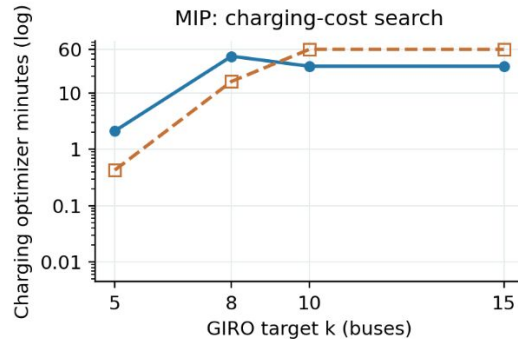
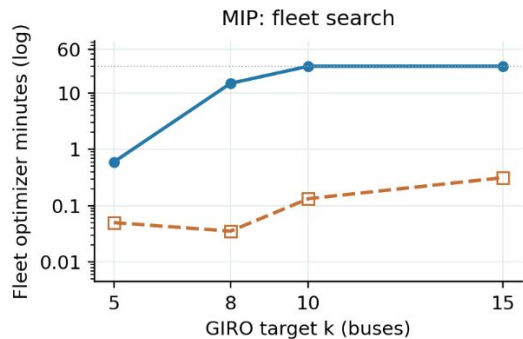
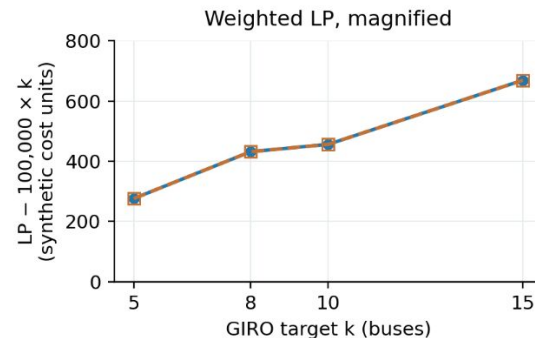
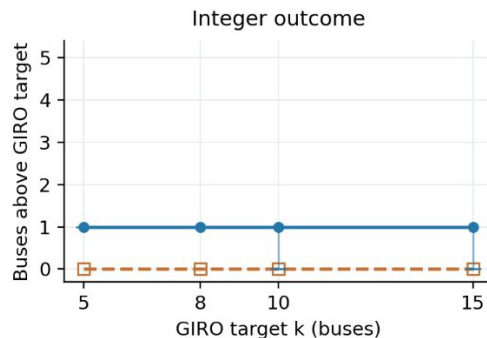
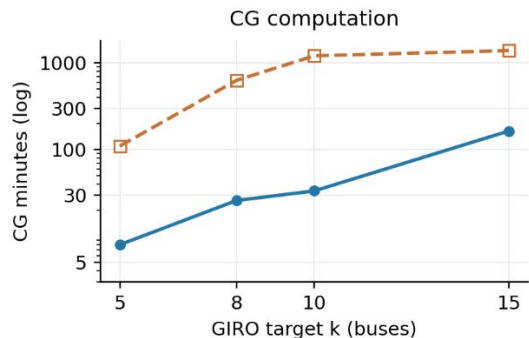
Chain 4: computation and solution quality

—●— Fresh: direct CG / final pool MIP —□— Sequential: cumulative CG / final pool MIP



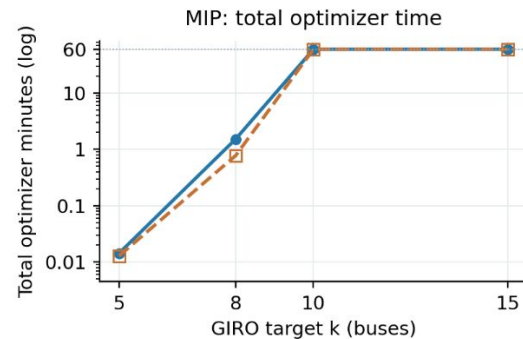
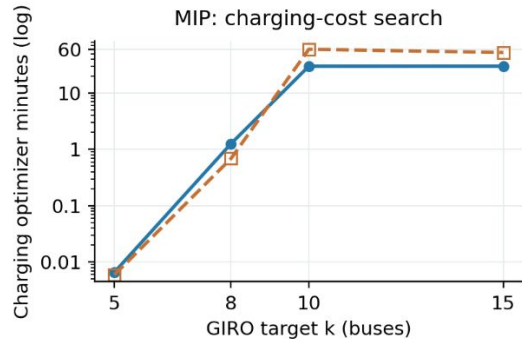
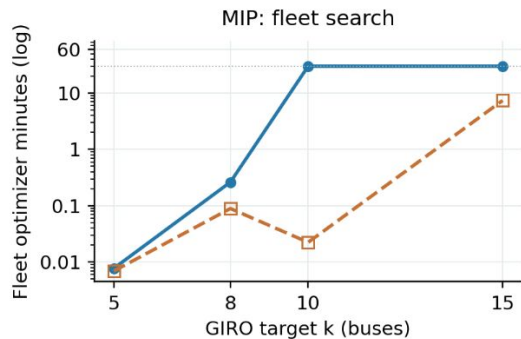
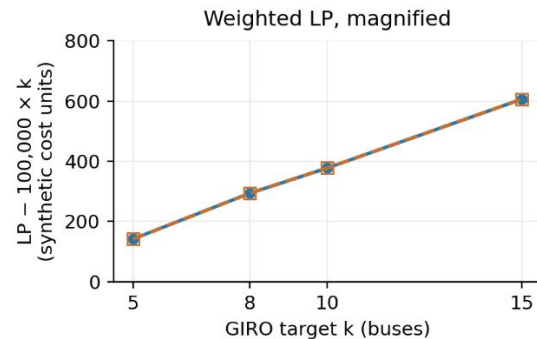
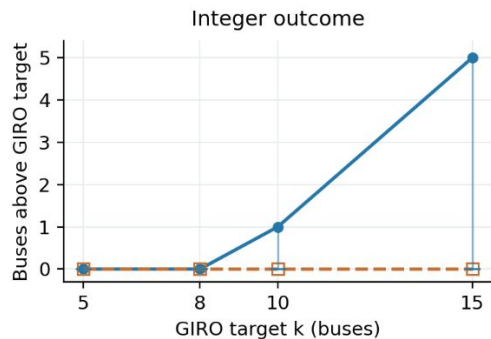
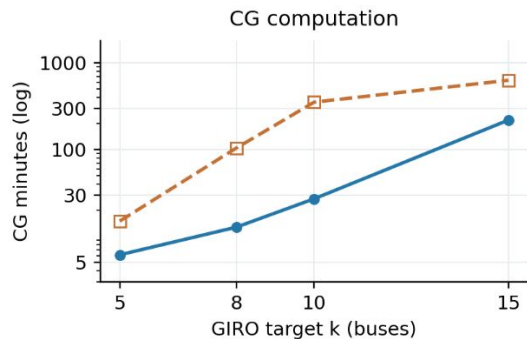
Chain 5: computation and solution quality

—●— Fresh: direct CG / final pool MIP —□— Sequential: cumulative CG / final pool MIP



Chain 6: computation and solution quality

—●— Fresh: direct CG / final pool MIP —□— Sequential: cumulative CG / final pool MIP



Fleet search is faster; charging uses the remaining time

Chain 1, $k=8$: fresh proves 9 buses in 26.3 minutes.

Sequential proves 8 buses in 27 seconds.

Both then run charging optimization until nearly one hour.

Across 24 cases: median fleet search is 30 minutes fresh,
versus 5.31 seconds sequential. All sequential fleets are proved.

Fresh $k=5$ is already fast: every MIP finishes within 3 minutes.

236.44 kWh: all 195 failures can be repaired

194 routes need only a charging-amount adjustment inside the original intervals. One needs a 0.15-second extension.

All 487 route occurrences pass independent replay.

Trips, bus counts, charging stops and start counts stay fixed.

Median net extra energy per repaired route: 2.02 kWh.

Capacity-only test: 240 kW, zero reserve, no shared capacity.

Continuous repair establishes feasibility, not a new LP proof.

Integer-directed pricing: how the search works

Start from the fresh pool and impose the target fleet cap $K = 8$.

Fix one promising fractional route at $x_r = 1$.

Re-solve the LP. Remaining trips and bus slots change the duals.

Price new routes that help complete this partial integer choice.

Fix another route, or release a fixing and try an alternative.

Keep new columns and pass any discovered cover to the final MIP.

No known sequential routes enter this test. The search is heuristic.

Why a useful integer route can have positive reduced cost

Covering LP: $\min c^T x$, subject to $Ax \geq 1$ and $x \geq 0$.

Dual: $\max 1^T \pi$, subject to $A^T \pi \leq c$ and $\pi \geq 0$.

For any integer cover y : $c^T y - z_{LP} = r^T y + \pi^T (Ay - 1)$,

where $r = c - A^T \pi$ is the vector of route reduced costs.

C1's eight-route witness: $96.272 = 72.142 + 24.130$.

The integer cover pays a small premium above the fractional LP.

Those positive reduced-cost routes can still reduce the integer fleet.

Fixing routes changes the pricing problem

For fixed routes F , free routes must cover $1 - A_F^T \pi$,
using at most $K - |F|$ remaining bus slots.

New reduced cost: $r_r = c_r - a_r^T \pi - \mu$, with $\mu \leq 0$.

π prices uncovered service; μ prices the remaining fleet cap.

A formerly positive reduced-cost route can now have negative reduced cost.

The search tries up to 3 alternatives per level and 2 restarts.

It is a bounded diving heuristic, not exhaustive branch-and-price.

Paired k=8 test: seven recoveries versus zero

Chain	New pricing hits	Unchanged-pool hits
C1	2 / 2	0 / 2
C3	2 / 2	0 / 2
C4	1 / 2	0 / 2
C5	2 / 2	0 / 2

All seven hits: 8 buses and finite-pool bound 8.

Treatment includes new columns and its own incumbent start.

Four selected instances, each repeated with two solver seeds.

What the new pricing test establishes

The method generated useful routes without importing known covers.

Five controls proved nine buses in their unchanged pools.

The treatment's only miss ended at nine with bound eight, still open.

Each pair shares a nominal one-hour dive-plus-solver budget.

Seven successful runs took 14.1–35.4 minutes including setup.

Physics remains 240 kWh / 240 kW, with no shared capacity.

Duplicate service remains in six hits. Global optimality is unproved.